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Why Do Firms Borrow Directly from Nonbanks?

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1 Big picture

- **Question.** Why do middle-market firms borrow directly from nonbank financial intermediaries instead of commercial banks, and what loan terms do they receive when they do so?
- **Setting.** The paper studies publicly traded U.S. middle-market firms during 2010–2015, using hand-collected credit agreements filed with the SEC.
- **Key market.** Middle-market firms are important because they are large enough to disclose detailed credit agreements but often too small or opaque to rely heavily on public debt markets.
- **Main empirical fact.** Direct nonbank lending is widespread: nonbanks originate about one-third of all loans in the sample.
- **Main explanation.** Bank regulation constrains banks' willingness to lend to two borrower groups: firms with negative EBITDA and firms with debt/EBITDA above six.
- **Baseline finding.** Firms with negative EBITDA are roughly 32 percentage points more likely to borrow from nonbanks after controls; firms with debt/EBITDA above six are about 15 percentage points more likely to do so.
- **Economic magnitude.** Since the unconditional probability of nonbank borrowing is about 32%, the negative EBITDA effect is very large.
- **Regulatory channel.** The paper links the result to bank supervisory rules that classify or scrutinize loans to unprofitable and highly levered firms, especially cash-flow-based loans.
- **Local-market evidence.** If a firm is located in a local banking market dominated by OCC-supervised banks, negative EBITDA has a much larger effect on the probability of borrowing from a nonbank.
- **Loan pricing.** Nonbank loans are much more expensive in the raw data. After controlling for borrower characteristics, loan terms, and distance, the nonbank premium remains economically large for constrained borrowers.
- **Nonprice terms.** Nonbank loans are less likely to have financial covenants and performance pricing, more likely to include warrants or convertibility, more likely to be fixed-rate, and more likely to occupy a riskier capital-structure position.
- **RDD evidence.** Around the zero-EBITDA boundary, negative EBITDA produces a first-stage jump in nonbank borrowing. A fuzzy RDD implies that nonbank borrowing raises interest rates and changes covenant and maturity terms for marginal cash-flow borrowers.
- **Future performance.** After controlling for observables, nonbank borrowers do not appear to perform worse ex post, suggesting that nonbanks are not simply selecting unobservably worse firms or charging high rates only to compensate for expected losses.
- **Takeaway.** The paper shows that bank regulation can segment the commercial credit market, pushing risky but potentially viable firms toward less-regulated lenders that provide different and more expensive debt contracts.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper focuses on middle-market firms because they are information-sensitive borrowers that traditionally rely on monitored private debt.
- These firms are not the largest public firms with easy access to bond markets or broadly syndicated loans.
- At the same time, because they are public companies, they file credit agreements and financial statements with the SEC.
- This combination allows the authors to observe both borrower characteristics and detailed contract terms.

Interpretation. The setting is useful because it reveals a segment of direct lending that is usually missing from DealScan and other standard loan databases.

2.2 Sample construction

- The authors begin with a random sample of 750 publicly traded U.S. middle-market firms.
- Middle-market firms are defined as firms with revenue between \$10 million and \$1 billion.
- The sample period is 2010–2015.
- The final loan sample contains 1,269 debt contracts entered into by 579 borrowers.
- The authors exclude financial firms, utilities, foreign firms, ADRs, and firms with very low leverage.
- Loan agreements are obtained mainly from SEC filings through Capital IQ and supplemented with DealScan.
- The paper keeps original contracts and amended-and-restated agreements but excludes simple amendments, waivers, joinders, bonds, and nonfinancial lenders.

Interpretation. The data are labor intensive but valuable: many direct nonbank loans, especially loans from hedge funds, private equity, venture capital, and asset managers, are not covered by DealScan.

2.3 Bank versus nonbank lender classification

Lenders are classified into categories such as:

- commercial banks,
- bank-affiliated finance companies,
- unaffiliated finance companies,

- investment banks,
- insurance companies,
- business development companies,
- private equity or venture capital funds,
- hedge funds,
- investment managers,
- other nonbank financial intermediaries.

The main dependent variable in many specifications is:

$$Nonbank_{ijt} = \mathbf{1}\{\text{loan } j \text{ to firm } i \text{ is originated by a nonbank lender at time } t\}. \quad (1)$$

Interpretation. The key object is not whether a nonbank participates in a bank-led syndicate. It is whether the nonbank negotiates directly with the borrower and originates the loan.

2.4 Borrower characteristics

Borrower characteristics are measured as of the quarter before loan origination. Important variables include:

- total assets,
- EBITDA,
- negative EBITDA dummy,
- debt/EBITDA,
- debt/EBITDA greater than six dummy,
- profitability,
- leverage and change in leverage,
- absolute asset growth,
- market-to-book,
- R&D intensity,
- PP&E,
- current ratio,
- firm age,
- sales growth,
- stock-return volatility,
- past stock return.

The two central regulatory variables are:

$$\mathbf{1}\{EBITDA_{it} < 0\} \quad (2)$$

and

$$\mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\}. \quad (3)$$

Interpretation. Negative EBITDA is the cleanest proxy for borrowers whose primary repayment source is weak. Debt/EBITDA above six captures highly levered borrowers targeted by leveraged lending guidance.

2.5 Loan characteristics

The authors hand-code detailed contract terms, including:

- loan amount,
- maturity,
- fixed versus floating rate,
- initial interest rate,
- up-front and annual fees,
- seniority,
- security,
- second-lien status,
- financial covenants,
- performance pricing,
- warrants,
- convertibility,
- tranche structure.

For a floating-rate loan, the initial interest rate is measured as the relevant base rate, usually LIBOR, plus the spread. For fixed-rate loans, it is the fixed stated rate.

Interpretation. This is important because lender choice may affect not only the price of debt but also monitoring, flexibility, and incentive alignment.

2.6 Local banking market variables

A firm's local banking market is defined as its MSA or non-MSA county. The key local-market variable is:

$$OCCShare_m = \frac{\text{Deposits of OCC-supervised commercial-lending banks in market } m}{\text{Deposits of all commercial-lending banks in market } m}. \quad (4)$$

Interpretation. OCC share proxies for how much of the local bank credit supply is supervised by a stricter federal regulator. If regulation is the mechanism, negative EBITDA firms should be more likely to use nonbanks in markets with high OCC share.

2.7 Descriptive facts

- Nonbanks originate 410 of 1,269 loans, or about 32% of loans.
- The largest nonbank categories include finance companies, private equity/venture capital funds, and hedge funds.
- Only about 19% of nonbank loans are tracked in DealScan.
- Nonbank borrowers are smaller, younger, more volatile, more levered, and more likely to have negative EBITDA.
- In raw differences, nonbank loans have higher interest rates, larger up-front fees, fewer financial covenants, fewer performance-pricing provisions, and more warrants and convertibility features.

Interpretation. Nonbank borrowing is not a tiny corner of the market. It is a large and distinct source of credit for riskier and more constrained middle-market firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework

The paper uses a search-and-bargaining intuition.

- Firms need loans and search for potential lenders one at a time.
- Lenders screen borrowers and make offers that include both price and nonprice terms.
- Banks have cheaper deposit funding and monitoring expertise but face regulatory constraints.
- Nonbanks do not face the same bank regulations but may have higher funding costs and different monitoring technology.
- Firms often begin with local banks because search and information costs are lower.
- If local banks are constrained or unwilling to lend, firms expand their search to nonlocal banks and nonbanks.

Key friction. Search is costly. Once a lender knows that a borrower has few attractive alternatives, the lender has more bargaining power.

Regulatory prediction. If bank regulation makes loans to negative EBITDA or highly levered firms costly for banks, these borrowers face a reduced set of bank lenders and are more likely to borrow from nonbanks.

Pricing prediction. Because these borrowers have higher effective search costs, nonbank lenders can charge higher interest rates, especially when banks are less willing to compete.

3.2 Baseline probability of borrowing from a nonbank

The baseline empirical model is a linear probability model:

$$\begin{aligned} Nonbank_{ijt} = & \alpha + \beta_1 \mathbf{1}\{EBITDA_{it} < 0\} + \beta_2 \frac{Debt_{it}}{EBITDA_{it}} + \beta_3 \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \\ & + \Gamma' X_{it} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \end{aligned} \quad (5)$$

where:

- $Nonbank_{ijt}$ equals one if loan j to borrower i at time t is originated by a nonbank;
- X_{it} includes borrower controls such as size, leverage, R&D, current ratio, firm age, market-to-book, sales growth, volatility, and past returns;
- δ_t are year fixed effects;
- $\delta_{k(i)}$ are industry fixed effects.

Interpretation. The coefficients β_1 and β_3 test whether regulatory flags—negative EBITDA and debt/EBITDA above six—predict nonbank borrowing after controlling for ordinary risk characteristics.

3.3 Asset-based loan exception

A central distinction is between cash-flow loans and asset-based loans. For cash-flow loans, operating cash flow is the primary repayment source. For asset-based loans, collateral such as receivables and inventory is more central.

The paper therefore estimates specifications with an asset-based loan interaction:

$$\begin{aligned} Nonbank_{ijt} = & \alpha + \beta_1 \mathbf{1}\{EBITDA_{it} < 0\} + \beta_2 ABL_{ijt} + \beta_3 \mathbf{1}\{EBITDA_{it} < 0\} \times ABL_{ijt} \\ & + \Gamma' X_{it} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \end{aligned} \quad (6)$$

Interpretation. If bank regulation is especially about cash-flow repayment capacity, the negative EBITDA effect should be weaker for asset-based loans. This is exactly the pattern the paper emphasizes.

3.4 Post-guidance interaction

The paper also studies the 2013 Interagency Leveraged Loan Guidance. Define:

$$PostGuidance_t = \mathbf{1}\{t \geq \text{May 21, 2013}\}. \quad (7)$$

A representative specification includes interactions such as:

$$\begin{aligned} Nonbank_{ijt} = & \alpha + \beta_1 \mathbf{1}\{EBITDA_{it} < 0\} + \beta_2 \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \\ & + \beta_3 \mathbf{1}\{EBITDA_{it} < 0\} \times PostGuidance_t \\ & + \beta_4 \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \times PostGuidance_t \\ & + \Gamma' X_{it} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \end{aligned} \quad (8)$$

Interpretation. These terms ask whether regulatory constraints became more binding after revised leveraged lending guidance.

3.5 Borrowing from stricter versus more lenient bank regulators

Conditional on borrowing from a bank, the paper studies whether risky borrowers are less likely to borrow from OCC-supervised banks:

$$\begin{aligned} OCCBank_{ijt} = & \alpha + \beta_1 \mathbf{1}\{EBITDA_{it} < 0\} + \beta_2 \frac{Debt_{it}}{EBITDA_{it}} + \beta_3 \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \\ & + \Gamma' X_{it} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \end{aligned} \quad (9)$$

Interpretation. A negative β_1 means that negative EBITDA firms that still borrow from banks are less likely to use nationally chartered OCC-supervised banks, consistent with stricter supervision reducing risky bank lending.

3.6 Local banking market specification

The local-market design asks whether a firm's probability of nonbank borrowing depends on how much of its nearby banking market is OCC-supervised:

$$\begin{aligned} Nonbank_{ijt} = & \alpha + \beta_1 \mathbf{1}\{EBITDA_{it} < 0\} + \beta_2 OCCShare_{m(i)} \\ & + \beta_3 \mathbf{1}\{EBITDA_{it} < 0\} \times OCCShare_{m(i)} \\ & + \beta_4 \frac{Debt_{it}}{EBITDA_{it}} \times OCCShare_{m(i)} \\ & + \beta_5 \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \times OCCShare_{m(i)} \\ & + \Gamma' X_{it} + \Lambda' Z_{m(i)t} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \end{aligned} \quad (10)$$

where $Z_{m(i)t}$ includes local economic controls such as unemployment, income, population, deposits, and bank capital.

Interpretation. The coefficient β_3 is the key. If local OCC share is high, negative EBITDA firms should have fewer viable bank options and therefore be more likely to borrow from nonbanks.

3.7 Interest rate regressions

The loan pricing model is:

$$Rate_{ijt} = \alpha + \theta Nonbank_{ijt} + \Gamma' X_{it} + \Phi' L_{ijt} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}, \quad (11)$$

where:

- $Rate_{ijt}$ is the initial interest rate in basis points;
- X_{it} contains borrower controls;
- L_{ijt} contains other loan terms such as amount, maturity, security, seniority, warrants, convertibility, covenants, performance pricing, fixed-rate status, and distance.

A richer version interacts lender type with regulatory-risk variables:

$$\begin{aligned} Rate_{ijt} = & \alpha + \theta_0 Nonbank_{ijt} + \theta_1 Nonbank_{ijt} \times \mathbf{1}\{EBITDA_{it} < 0\} \\ & + \theta_2 Nonbank_{ijt} \times \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \\ & + \theta_3 PostGuidance_t \times \mathbf{1}\{EBITDA_{it} < 0\} \\ & + \theta_4 PostGuidance_t \times \mathbf{1}\left\{\frac{Debt_{it}}{EBITDA_{it}} > 6\right\} \\ & + \Gamma' X_{it} + \Phi' L_{ijt} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \end{aligned} \quad (12)$$

Interpretation. This specification distinguishes an overall nonbank premium from a special premium charged to borrowers for whom banks are regulatory-constrained competitors.

3.8 Nonprice loan terms

For each nonprice term Y_{ijt}^k , the paper estimates:

$$Y_{ijt}^k = \alpha_k + \theta_k Nonbank_{ijt} + \Gamma'_k X_{it} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}^k. \quad (13)$$

Examples of Y^k include:

- log loan amount,
- maturity,
- seniority,
- security,
- financial covenants,
- performance pricing,
- warrants,
- convertibility,

- fixed-rate status,
- up-front fees,
- annual fees,
- second-lien status.

Interpretation. These regressions ask whether nonbanks lend through a different contract technology, not merely whether they charge higher interest rates.

3.9 Fuzzy regression discontinuity around zero EBITDA

The strongest identification exercise uses the discontinuity at zero EBITDA. The treatment is non-bank borrowing:

$$D_i = \text{Nonbank}_i. \quad (14)$$

The running variable is trailing twelve-month EBITDA:

$$R_i = \text{EBITDA}_i. \quad (15)$$

The first stage is:

$$D_i = \pi_0 + \pi_1 \mathbf{1}\{R_i < 0\} + f_-(R_i) \mathbf{1}\{R_i < 0\} + f_+(R_i) \mathbf{1}\{R_i \geq 0\} + \Gamma' X_i + u_i. \quad (16)$$

The second stage is:

$$Y_i = \alpha + \tau \widehat{D}_i + g_-(R_i) \mathbf{1}\{R_i < 0\} + g_+(R_i) \mathbf{1}\{R_i \geq 0\} + \Gamma' X_i + \nu_i. \quad (17)$$

Interpretation. Negative EBITDA does not perfectly determine nonbank borrowing, so the design is fuzzy rather than sharp. The coefficient τ estimates the treatment effect of nonbank borrowing for marginal cash-flow borrowers near the zero-EBITDA regulatory boundary.

3.10 Future performance tests

The paper estimates bankruptcy and subsequent-performance regressions such as:

$$\text{Bankruptcy}_{i,t+5} = \alpha + \theta \text{Nonbank}_{ijt} + \Gamma' X_{it} + \delta_t + \delta_{k(i)} + \varepsilon_{ijt}. \quad (18)$$

Interpretation. If nonbanks are merely lending to unobservably worse firms or charging high rates to compensate for expected default losses, nonbank borrowers should perform worse after loan origination. The paper does not find this pattern after controls.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline effect

The baseline regressions compare loans to firms with similar observable characteristics but different EBITDA signs, leverage ratios, and lender types.

- Year fixed effects absorb aggregate changes in credit conditions.
- Industry fixed effects absorb broad differences in collateral, business models, and financing needs.
- Rich controls absorb observable credit risk and borrower characteristics.
- Some specifications include borrower fixed effects, using within-firm variation over time.

Interpretation. The baseline design is not enough by itself to prove causality, but it documents that negative EBITDA and high debt/EBITDA are large predictors of nonbank borrowing even after standard risk controls.

4.2 Regulatory thresholds as quasi-experimental variation

The empirical logic relies on specific regulatory thresholds:

- Negative EBITDA matters because bank supervisors focus on operating cash flow as a primary repayment source.
- Debt/EBITDA above six matters because leveraged lending guidance treats very high leverage as a supervisory concern.

Interpretation. These thresholds are not arbitrary risk proxies chosen after looking at the data. They are grounded in bank supervisory rules and industry practice.

4.3 Cash-flow loans versus asset-based loans

A useful falsification-style comparison is between cash-flow loans and asset-based loans.

- For cash-flow loans, negative EBITDA should trigger regulatory concern.
- For asset-based loans, collateral rather than operating cash flow is the repayment focus.
- The negative EBITDA effect is therefore expected to be stronger for cash-flow loans.

Interpretation. This comparison strengthens the regulatory interpretation because the same EBITDA sign has different relevance across loan technologies.

4.4 OCC-supervised banks and local banking markets

The paper uses variation in regulator strictness.

- OCC-supervised banks are treated as more strictly supervised than many state-chartered banks.
- Conditional on borrowing from a bank, negative EBITDA firms are less likely to borrow from OCC-supervised banks.
- Negative EBITDA firms in markets with high OCC deposit share are more likely to borrow from nonbanks.

Interpretation. This is a powerful part of the paper because it connects lender choice to the regulatory environment of the borrower's nearby bank supply.

4.5 Fuzzy RDD around zero EBITDA

The fuzzy RDD relies on a discontinuous change in bank-lending constraints at zero EBITDA.

Credibility checks include:

- a visible jump in nonbank borrowing at zero EBITDA;
- local polynomial first-stage estimates around the cutoff;
- smoothness of the EBITDA distribution around zero;
- no consistent discontinuity in other borrower covariates;
- focus on cash-flow loans, where the regulatory logic is strongest.

Interpretation. The RDD asks whether borrowers just below zero EBITDA, who are pushed toward nonbanks, receive different terms than otherwise similar borrowers just above zero EBITDA.

4.6 Addressing alternative explanations

- **Riskier borrowers.** The paper controls for many risk measures and shows that nonbank borrowers do not perform worse ex post after controls.
- **Different contract bundle.** The interest rate results remain meaningful after controlling for many nonprice loan terms.
- **Distance and search costs.** Borrower-lender distance matters, and including distance reduces the nonbank premium, consistent with search-cost frictions.
- **Unobservable borrower quality.** The RDD and future performance tests help address this concern, though they cannot eliminate it entirely.
- **Nonbank funding costs.** Higher funding costs could explain a general nonbank premium but not the especially large premium for negative EBITDA borrowers.

4.7 What the paper can and cannot distinguish

- The paper is strong on the claim that bank regulation shifts some borrowers toward nonbanks.
- It is also strong on the claim that nonbank loans have different price and nonprice terms.

- The paper cannot observe all rejected loan applications or every lender contacted during search.
- It cannot fully decompose the interest-rate premium into funding-cost, monitoring-cost, market-power, and search-cost components.
- It cannot observe the full set of unfiled private credit arrangements for nonpublic firms.

Interpretation. The paper’s central contribution is not a randomized experiment on all credit supply. It is a carefully triangulated empirical case that regulation-driven market segmentation is a major force in direct nonbank lending.

5 Tables and figures

5.1 Table 1: Number of originated loans, lender types, and DealScan match rates

Read.

- The sample has 1,269 loans, with nonbanks originating 32.31%.
- Nonbank lending is present in every year from 2010 to 2015.
- Finance companies, PE/VC firms, and hedge funds are the largest nonbank groups.
- DealScan captures only 19.27% of nonbank loans.

Economic message. Standard loan databases miss a large part of direct nonbank lending. The hand-collected data are essential for studying this market.

Table 1
Number of originated loans, lender types, and DealScan match rates

A. Originated loans per year			
	Obs.		% nonbank
2010	233		32.19
2011	268		30.22
2012	244		33.61
2013	200		35.00
2014	202		29.70
2015	122		34.43
Total observations	1,269		32.31
B. Lender types and DealScan match rates			
	Obs.	% of nonbank deals	% tracked in DealScan
Bank	859		52.97
Nonbanks:			
Bank FCO	52	12.68	26.92
FCO	93	22.68	25.81
Investment bank	42	10.24	73.81
Insurance	22	5.37	4.55
BDC	17	4.15	11.76
PE/VC	78	19.02	0.00
Hedge fund	68	16.59	5.88
Investment manager	34	8.29	5.88
Other	4	0.98	25.00
Total observations	410	100.00	19.27

Panel A reports the total number of originated loans and the share of loans extended by nonbanks in each year. Panel B reports the number originated loans and the percentage included in the DealScan database for each nonbank lender type. The sample includes all borrowings of a random sample of 750 middle-market firms originated and filed with the SEC during the 2010–2015 period. We treat multiple tranches within a given package as a single observation.

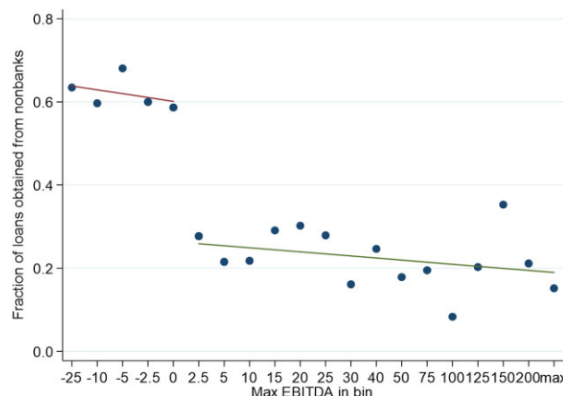


Figure 1
Fraction of loans obtained from nonbanks, by EBITDA bin
This figure graphs the fraction of loans obtained from nonbanks at different levels of EBITDA. Loans are allocated into 20 bins based on borrower’s trailing 12-month EBITDA at loan origination. The x-axis represents the upper limit of EBITDA for each bin. The choice of bin limits roughly follows the distribution obtained by splitting EBITDA into 20 quantiles, rounded to multiples of five.

5.2 Figure 1: Fraction of loans obtained from nonbanks by EBITDA bin

Read.

- The fraction of loans from nonbanks is much higher for firms with negative EBITDA.
- The share drops sharply around the zero-EBITDA threshold.
- Positive-EBITDA borrowers rely much less on nonbanks.

Economic message. The visual pattern motivates the regulatory interpretation and the later RDD design.

5.3 Table 3: Probability of borrowing from a nonbank lender

Read.

- Negative EBITDA has a large positive coefficient across specifications.
- The effect is about 0.32 in the main controlled specification.
- Debt/EBITDA greater than six is also positively related to nonbank borrowing.
- The interaction with asset-based lending weakens the negative EBITDA effect, consistent with a cash-flow-loan regulatory mechanism.
- Borrower fixed effects preserve the main negative EBITDA result.

Economic message. The key borrower-lender matching result is that nonbanks serve borrowers that banks are reluctant or costly to fund under regulatory supervision.

	(1)	(2)	(3)	(4)	(5)
<i>EBITDA < 0</i>	0.39*** (7.78)	0.32*** (5.99)	0.47*** (7.59)	0.38*** (3.37)	0.31*** (2.71)
<i>Debt/EBITDA</i>	0.01 (1.06)	0.00 (0.07)	0.01 (1.05)	0.02 (0.81)	0.01 (0.30)
<i>Debt/EBITDA > 6x</i>	0.14* (1.75)	0.15* (1.86)	0.14 (1.36)	0.16 (0.98)	0.20 (1.16)
<i>EBITDA</i>	-0.00 (-0.92)	-0.00 (-0.73)	-0.00 (-0.65)	0.00 (0.06)	0.00 (0.13)
<i>ln(Assets)</i>	0.01 (0.49)	0.01 (0.92)	0.01 (0.83)	-0.03 (-0.42)	-0.06 (-0.68)
<i>Leverage</i>	0.22*** (3.40)	0.24*** (3.33)	0.17** (2.45)	0.22 (1.45)	0.20 (1.42)
<i>Δleverage</i>	0.37*** (4.20)	0.29*** (2.79)	0.24** (2.53)	0.30* (1.84)	0.33* (1.88)
<i>[Asset growth]</i>	0.16*** (5.20)	0.16*** (4.67)	0.13*** (3.81)	0.08 (1.46)	0.08 (1.24)
<i>Research expenses</i>	-0.03 (-0.24)	0.08 (0.54)	0.11 (0.88)	-0.17 (-0.39)	0.30 (0.52)
<i>PP&E</i>	-0.04 (-0.45)	-0.04 (-0.41)	-0.05 (-0.59)	-0.10 (-0.38)	-0.17 (-0.52)
<i>Current ratio</i>	-0.03*** (-3.15)	-0.03*** (-2.61)	-0.03*** (-2.87)	-0.02 (-1.43)	-0.02 (-1.04)
<i>ln(Firm age)</i>	-0.02 (-0.95)	-0.02 (-0.57)	-0.02 (-0.67)	-0.23 (-1.08)	-0.24 (-0.89)
<i>Market-to-book</i>		-0.02 (-1.41)	-0.02 (-1.56)	-0.03 (-1.00)	-0.03 (-1.09)
<i>Sales growth</i>		0.06* (1.65)	0.06* (1.72)	0.06* (1.23)	0.06 (1.23)
<i>Volatility</i>		0.06** (2.07)	0.05* (1.73)	0.06 (1.23)	0.06 (1.23)
<i>Past return</i>		-0.11*** (-3.23)	-0.10*** (-2.97)	-0.07 (-0.98)	-0.07 (-0.98)
<i>(EBITDA < 0) * ABL</i>			-0.35*** (-5.09)	-0.19* (-1.66)	-0.14 (-1.20)
<i>ABL</i>			-0.08** (-2.04)	-0.23*** (-3.09)	-0.22*** (-2.78)
<i>Debt/EBITDA *</i>			-0.02 (-1.15)	-0.01 (-0.34)	-0.00 (-0.16)
<i>Postguidance</i>			-0.02 (-0.13)	-0.17 (-0.78)	-0.21 (-0.91)
<i>(Debt/EBITDA > 6x) *</i>			0.02 (0.22)	0.04 (0.39)	0.06 (0.57)
<i>Postguidance</i>			0.14 (1.10)	1.24 (1.28)	1.42 (1.15)
Constant	0.16 (1.59)	0.10 (0.76)			
Year effects	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	No	No
Borrower effects	No	No	No	Yes	Yes
<i>N</i>	1,193	1,121	1,121	1,193	1,121
<i>R</i> ²	.25	.25	.30	.74	.74

This table reports the results from linear probability models to determine whether a loan is extended by a nonbank lender. The sample includes all nonbond borrowings of a random sample of 750 middle-market firms originated during the 2010–2015 period. Observations are aggregated at the deal level using the average value of each

	(1)	(2)	(3)	(4)
<i>EBITDA < 0</i>	-0.30*** (-3.76)	-0.38*** (-3.39)	-0.27* (-1.68)	-0.34 (-1.64)
<i>Debt/EBITDA</i>	-0.03* (-1.91)	-0.03* (-1.93)	-0.02 (-0.64)	-0.02 (-0.54)
<i>Debt/EBITDA > 6x</i>	0.01 (0.11)	0.10 (0.79)	-0.03 (-0.12)	0.01 (0.05)
<i>EBITDA</i>	0.00 (0.03)	0.00 (0.00)	0.00 (0.21)	0.00 (0.12)
<i>ln(Assets)</i>	0.10*** (5.12)	0.10*** (5.11)	-0.00 (-0.02)	-0.03 (-0.22)
<i>Leverage</i>	0.12 (0.86)	0.13 (0.95)	-0.14 (-0.33)	-0.21 (-0.46)
<i>Δleverage</i>	-0.12 (-0.62)	-0.12 (-0.65)	-0.28 (-0.68)	-0.35 (-0.80)
<i>[Asset growth]</i>	0.08 (1.19)	0.08 (1.27)	0.10 (0.96)	0.11 (1.03)
<i>Research expenses</i>	-0.51** (-2.05)	-0.57** (-2.28)	-2.08 (-1.56)	-1.97 (-1.52)
<i>PP&E</i>	-0.22** (-2.18)	-0.24** (-2.45)	0.52 (0.76)	0.52 (0.76)
<i>Current ratio</i>	0.00 (0.32)	0.00 (0.32)	0.02 (0.83)	0.02 (0.80)
<i>ln(Firm age)</i>	0.00 (0.08)	0.00 (0.15)	0.25 (0.66)	0.36 (0.94)
<i>Market-to-book</i>	0.03 (1.59)	0.03 (1.50)	0.02 (0.41)	0.02 (0.30)
<i>Sales growth</i>	-0.11* (-1.71)	-0.11* (-1.73)	-0.08 (-0.58)	-0.08 (-0.67)
<i>Volatility</i>	0.05 (1.42)	0.05 (1.41)	0.08 (1.57)	0.09* (1.77)
<i>Past return</i>	-0.01 (-0.21)	-0.01 (-0.15)	-0.02 (-0.20)	-0.03 (-0.29)
<i>(EBITDA < 0) * ABL</i>		0.27** (2.35)		0.22 (1.13)
<i>ABL</i>		-0.06 (-1.34)		-0.08 (-0.78)
<i>Debt/EBITDA *</i>		0.01 (0.52)		-0.00 (-0.02)
<i>Postguidance</i>		-0.19 (-0.88)		-0.08 (-0.18)
<i>(Debt/EBITDA > 6x) *</i>		-0.23* (-1.89)		-0.16 (-0.74)
<i>Postguidance</i>		0.02 (0.22)		0.11 (0.66)
Constant	0.26* (1.74)	0.30** (2.03)	-0.30 (-0.15)	-0.70 (-0.34)
Year effects	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	No	No
Borrower effects	No	No	Yes	Yes
<i>N</i>	719	719	719	719
<i>R</i> ²	.26	.28	.83	.84

This table reports the results from linear probability models where the dependent variable equals one if the

5.4 Table 4: Probability of borrowing from an OCC-regulated bank

Read.

- Conditional on borrowing from a bank, negative EBITDA firms are less likely to borrow from OCC-regulated banks.
- The negative EBITDA coefficient is large and negative in the main specifications.
- After the revised leveraged lending guidance, negative EBITDA firms become even less likely to borrow from OCC banks.

Economic message. The paper does not merely compare banks with nonbanks. It also shows that stricter bank regulators matter within the banking sector.

5.5 Table 5: Nonbank lending and leniency of regulators supervising local banks

Read.

- For cash-flow loans, the interaction between negative EBITDA and local OCC share is positive and significant.
- The interaction is not positive for asset-based loans.
- The local OCC-share effect is strongest for smaller firms.

Economic message. Firms with negative EBITDA are especially likely to turn to nonbanks when their nearby bank market is dominated by stricter regulators. This strongly supports the bank-regulation channel.

5.6 Table 6: Initial interest rate charged on bank versus nonbank loans

Read.

- The raw nonbank interest-rate premium is about 435 basis points.
- With borrower controls, the premium falls to about 305 basis points.
- With loan-term controls, it falls further to about 193 basis points.
- With distance controls, it is about 167 basis points.
- The interaction between nonbank lending and negative EBITDA is very large, around 254 basis points in the interacted specification.

Economic message. Nonbanks do not appear to win borrowers by offering cheaper debt. The premium is concentrated among borrowers for whom banks are constrained competitors.

Table 5

Nonbank lending and leniency of regulators supervising local banks

	Cash flow loans		Asset-based loans		Small firms		Large firms	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>EBITDA < 0</i>	0.19 (1.64)	0.17 (1.55)	0.31** (2.48)	0.29** (2.24)	0.00 (0.03)	0.00 (0.02)	0.45* (1.75)	0.46* (1.73)
<i>EBITDA < 0 *</i>	0.60*** (2.80)	0.60*** (2.91)	-0.25 (-0.93)	-0.22 (-0.79)	0.81*** (3.45)	0.83*** (3.58)	-0.58 (-1.29)	-0.65 (-1.37)
<i>OCC share</i>	-0.09 (-0.89)	-0.04 (-0.39)	0.26 (1.21)	0.30 (1.38)	-0.30** (-2.04)	-0.27* (-1.71)	0.10 (1.05)	0.16 (1.52)
<i>Debt/EBITDA > 6x</i>	0.24 (1.15)	0.25 (1.18)	-0.45 (-1.19)	-0.47 (-1.17)	0.31 (1.24)	0.39 (1.48)	0.17 (0.69)	0.11 (0.42)
<i>Debt/EBITDA > 6x *</i>	-0.07 (-0.15)	-0.08 (-0.17)	0.82 (1.13)	0.88 (1.14)	-0.54 (-0.98)	-0.71 (-1.18)	-0.17 (-0.36)	0.00 (0.00)
<i>OCC share</i>	-0.01 (-0.07)	-0.01 (-0.05)	0.10** (2.30)	0.10** (2.25)	-0.05 (-1.60)	-0.06 (-1.75)	0.04 (1.58)	0.05* (1.78)
<i>Debt/EBITDA *</i>	0.03 (0.76)	0.03 (0.60)	-0.17* (-1.91)	-0.18* (-1.92)	0.13** (2.05)	0.14** (2.14)	-0.06 (-1.38)	-0.08* (-1.82)
<i>Unemployment</i>	-0.01 (-0.50)	-0.01 (-0.50)	0.01 (0.28)	0.01 (0.28)	-0.01 (-0.45)	-0.01 (-0.45)	-0.01 (-0.52)	-0.01 (-0.52)
<i>Per capita income growth</i>	-0.01 (-0.01)	-0.01 (-0.01)	0.40 (0.39)	0.40 (0.39)	-0.27 (-0.34)	-0.27 (-0.34)	0.11 (0.14)	0.11 (0.14)
<i>ln(Per capita income)</i>	0.03 (0.23)	0.03 (0.23)	0.02 (0.13)	0.02 (0.13)	-0.09 (-0.75)	-0.09 (-0.75)	0.12 (0.86)	0.12 (0.86)
<i>ln(Population)</i>	0.11** (2.14)	0.11** (2.14)	-0.06 (-0.81)	-0.06 (-0.81)	0.04 (0.61)	0.04 (0.61)	0.04 (0.79)	0.04 (0.79)
<i>Tier 1 leverage ratio</i>	-0.03 (-0.83)	-0.03 (-0.83)	-0.04 (-0.84)	-0.04 (-0.84)	-0.07 (-1.57)	-0.07 (-1.57)	-0.03 (-0.68)	-0.03 (-0.68)
<i>ln(Deposits)</i>	-0.11** (-2.57)	-0.11** (-2.57)	0.06 (0.85)	0.06 (0.85)	-0.01 (-0.20)	-0.01 (-0.20)	0.06 (1.53)	0.06 (1.53)
<i>EBITDA</i>	-0.00* (-1.88)	-0.00* (-1.90)	0.00 (0.38)	0.00 (0.38)	-0.00 (-0.27)	-0.00 (-0.27)	-0.00 (-0.16)	-0.00 (-0.16)
<i>ln(Assets)</i>	0.01 (0.43)	0.01 (0.40)	0.01 (0.38)	0.01 (0.42)	0.02 (0.97)	0.02 (0.97)	0.01 (0.16)	0.00 (0.12)
<i>Leverage</i>	0.17** (2.27)	0.17** (2.38)	0.30** (2.06)	0.29* (1.92)	0.31*** (3.91)	0.32*** (4.06)	-0.02 (-0.15)	-0.00 (-0.00)
<i>Δleverage</i>	0.26** (2.43)	0.25** (2.29)	0.04 (0.19)	0.05 (0.23)	0.22* (1.74)	0.24* (1.92)	0.33 (1.17)	0.33 (1.19)
<i>ln(Assets growth)</i>	0.11*** (2.89)	0.11*** (2.85)	0.13 (1.25)	0.14 (1.39)	0.14*** (4.12)	0.15*** (4.04)	0.15 (1.42)	0.15 (1.35)
<i>Research expenses</i>	0.40*** (2.99)	0.38*** (2.90)	-0.30 (-1.59)	-0.29 (-1.50)	0.09 (0.59)	0.08 (0.56)	-0.01 (-0.02)	0.08 (0.21)
<i>PP&E</i>	0.03 (0.30)	0.04 (0.45)	-0.11 (-0.67)	-0.10 (-0.56)	-0.05 (-0.34)	-0.05 (-0.31)	-0.03 (-0.32)	-0.05 (-0.44)
<i>Current ratio</i>	-0.03** (-2.24)	-0.02** (-2.22)	-0.03** (-2.28)	-0.03** (-2.28)	-0.02 (-1.93)	-0.01 (-1.41)	-0.03** (-2.43)	-0.03** (-2.30)
<i>ln(Firm age)</i>	-0.03 (-0.79)	-0.02 (-0.66)	-0.01 (-0.41)	-0.01 (-0.36)	0.01 (0.35)	0.02 (0.62)	-0.05 (-1.14)	-0.04 (-1.10)
<i>Market-to-book</i>	-0.03** (-2.10)	-0.03** (-2.11)	0.01 (0.50)	0.01 (0.55)	-0.01 (-0.70)	-0.01 (-0.58)	-0.02 (-1.20)	-0.02 (-0.96)
<i>Sales growth</i>	0.09** (2.10)	0.08** (1.93)	0.03 (0.54)	0.03 (0.47)	0.06 (1.36)	0.06 (1.34)	0.18*** (2.95)	0.16** (2.72)
<i>Volatility</i>	0.05* (1.80)	0.06** (2.12)	-0.06 (-1.01)	-0.06 (-0.92)	0.07** (2.58)	0.07** (2.50)	0.30*** (2.85)	0.29** (2.72)
<i>Past return</i>	-0.09** (-2.16)	-0.09** (-2.32)	-0.14*** (-2.56)	-0.14*** (-2.63)	-0.15*** (-3.56)	-0.15*** (-3.66)	-0.06 (-1.11)	-0.05 (-1.04)
<i>Constant</i>	0.25 (1.49)	-0.21 (-0.12)	0.05 (-0.26)	0.20 (0.09)	0.05 (0.27)	0.13 (0.67)	0.12 (0.47)	-0.77 (-0.42)
<i>Year effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	775	775	343	343	559	559	560	560
<i>R²</i>	.36	.37	.24	.25	.34	.35	.20	.21

This table reports the results of linear probability models to determine whether a loan is extended by a nonbank lender on the share of OCC-supervised banks in the firm's local banking market: MSA or non-MSA county. The sample of banks consists of banks whose primary asset specialization is commercial lending, according to the *Commercial Bank of America* website.

Table 6

Initial interest rate charged on bank versus nonbank loans

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Nonbank</i>	434.82*** (10.66)	304.89*** (9.88)	192.87*** (6.68)	166.76*** (5.88)	55.41 (0.99)	234.45*** (3.69)	105.66 (1.14)
<i>EBITDA < 0</i>	192.41*** (5.77)	197.77*** (6.15)	184.34*** (5.62)	67.75** (2.23)	153.65*** (2.19)	133.24 (1.63)	
<i>Debt/EBITDA</i>	15.55** (2.32)	15.99*** (2.62)	13.20** (2.12)	16.80*** (2.89)	14.34 (1.33)	15.93 (1.20)	
<i>Debt/EBITDA > 6x</i>	-75.88 (-1.50)	-43.25 (-1.03)	-28.46 (-0.70)	-129.53*** (-2.79)	-23.74 (-0.38)	-112.38 (-1.09)	
<i>ln(Assets)</i>	-49.60*** (-5.63)	-45.39*** (-3.00)	-45.26*** (-2.91)	-38.29*** (-2.94)	-32.36 (-0.76)	-22.29 (-0.55)	
<i>EBITDA</i>	-0.02 (-0.62)	-0.02 (-0.68)	-0.04 (-0.97)	-0.02 (-0.61)	-0.04 (-0.48)	-0.04 (-0.47)	
<i>Leverage</i>	195.66*** (4.16)	115.33*** (2.92)	120.73*** (3.00)	114.74*** (2.89)	198.05** (2.04)	172.65* (1.66)	
<i>Δleverage</i>	333.83*** (5.13)	222.09*** (3.73)	225.88*** (3.69)	240.96*** (4.29)	329.59*** (2.44)	319.11*** (2.45)	
<i> Asset growth </i>	31.12 (1.37)	28.44 (1.59)	23.88 (1.29)	27.40* (1.72)	-18.30 (-0.41)	-5.53 (-0.13)	
<i>Research expenses</i>	29.53 (0.36)	-45.00 (-0.59)	-47.28 (-0.61)	-37.31 (-0.53)	523.95 (1.29)	409.44 (0.99)	
<i>PP&E</i>	-6.74 (-0.14)	5.98 (0.15)	0.25 (0.01)	-2.56 (-0.07)	-21.23 (-0.11)	-85.60 (-0.43)	
<i>Current ratio</i>	-2.35 (-0.45)	-3.22 (-0.71)	-2.87 (-0.64)	-0.70 (-0.16)	-15.31 (-0.17)	-20.00 (-1.53)	
<i>ln(Firm age)</i>	-8.02 (-0.59)	-11.41 (-0.98)	-14.05 (-0.92)	-47.67 (-1.28)	-14.11 (-0.46)	-14.11 (-0.12)	
<i>Market-to-book</i>	-8.67 (-1.17)	-8.31 (-1.17)	-6.85 (-1.30)	-6.73 (-1.35)	-8.98 (-0.37)	-10.25 (-0.43)	
<i>Sales growth</i>	32.44 (0.95)	38.84 (1.32)	36.37 (1.25)	34.84 (1.23)	-48.77 (-1.01)	-58.90 (-1.14)	
<i>Volatility</i>	-1.92 (-0.09)	4.85 (0.25)	4.48 (0.23)	7.79 (0.42)	-11.49 (-0.39)	-10.61 (-0.34)	
<i>Past return</i>	-52.22* (-1.83)	-69.54*** (-3.01)	-70.71*** (-3.03)	-65.52*** (-2.94)	-26.80 (-0.59)	-28.67 (-0.65)	
<i>ln(Amount)</i>	17.35 (1.53)	17.50 (1.48)	10.14 (1.04)	9.62 (0.44)	6.50 (0.30)		
<i>Performance pricing</i>	-55.00*** (-4.33)	-53.46*** (-4.38)	-56.62*** (-4.88)	-50.86 (-1.51)	-63.22* (-1.91)		
<i>Warrants</i>	94.49*** (2.23)	92.11** (2.21)	70.63* (1.69)	-15.56 (-0.10)	-10.94 (-0.07)		
<i>Convertible</i>	-248.63*** (-3.50)	-253.53*** (-3.38)	-288.57*** (-3.80)	-255.13 (-1.61)	-247.09 (-1.50)		
<i>Financial covenants</i>	-25.48 (-1.11)	-33.31 (-1.44)	-10.67 (-0.48)	0.98 (0.02)	22.94 (0.40)		
<i>Secured</i>	6.66 (0.30)	3.34 (0.15)	16.54 (0.85)	-13.95 (-0.24)	1.10 (0.02)		
<i>Second lien</i>	475.65*** (8.61)	481.86*** (8.78)	517.44*** (9.68)	342.77*** (3.23)	358.86*** (3.21)		
<i>Maturity</i>	-6.79 (-1.42)	-7.60* (-1.65)	-5.43 (-1.33)	6.19 (0.75)	6.28 (0.76)		
<i>Fixed-rate loan</i>	156.75*** (3.59)	152.08*** (3.53)	162.65*** (4.17)	118.23 (1.36)	144.51 (1.63)		
<i>Senior</i>	-208.79*** (-3.59)	-206.57*** (-3.47)	-197.53*** (-3.33)	-172.67 (-1.27)	-148.42 (-1.07)		
<i>ln(Distance)</i>	5.18 (1.46)	4.71 (1.37)	0.16 (0.16)	2.91 (0.30)			
<i>Same county</i>	-47.30*** (-3.43)	-52.75*** (-3.43)	-70.74 (-1.37)	-75.24 (-1.51)			
<i>EBITDA < 0 *</i>			254.31***	233.88**			

Table 7

Continued

	C: Other loan terms							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Fixed-rate loan	Fixed-rate loan	Up-front fee (bp)	Up-front fee (bp)	Annual fee (bp)	Annual fee (bp)	Second lien	Second lien
<i>Nonbank</i>	0.48*** (11.23)	0.46*** (7.10)	12.19* (1.76)	14.71 (1.00)	1.41 (0.60)	4.26 (0.91)	0.06*** (0.66)	0.17*** (2.75)
<i>ln(assets)</i>	-0.06*** (-1.18)	-0.06*** (-0.87)	-5.56*** (-1.72)	-17.68 (-1.41)	-0.28 (-0.41)	3.67 (1.00)	0.00 (0.21)	-0.01 (-0.80)
<i>EBITDA</i>	0.00* (1.67)	-0.00 (-0.00)	0.00 (0.29)	-0.00 (-0.05)	0.00 (0.22)	-0.00 (-0.73)	-0.00** (-1.43)	-0.00 (0.00)
<i>Debt/EBITDA</i>	-0.02** (-1.23)	-0.00 (-0.02)	1.65 (2.12)	0.32 (0.77)	0.48 (1.13)	0.61 (0.90)	0.01** (0.72)	0.00 (-0.09**)
<i>Debt/EBITDA > 6x</i>	0.01 (0.13)	-0.02 (-0.22)	2.12 (0.71)	15.51 (0.73)	-0.58 (-0.36)	0.04 (0.01)	-0.09** (-1.20)	0.00 (-2.03)
<i>EBITDA < 0</i>	-0.01 (-0.16)	0.00 (0.54)	30.60*** (3.69)	-4.92 (-0.26)	8.37*** (2.66)	6.18 (0.92)	-0.02 (-0.99)	-0.00 (-1.10)
<i>Leverage</i>	0.12** (2.05)	0.11 (1.01)	15.71 (0.78)	14.84 (0.47)	-5.54*** (-1.21)	0.02 (0.76)	0.00 (0.34)	0.00 (0.34)
<i>N</i>	1,089	1,089	956	956	959	959	1,128	1,128
<i>R²</i>	.43	.79	.12	.66	.05	.77	.05	.48
<i>Additional controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Firm effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table reports the results of OLS regressions of mortgage terms on lender type indicators, as well as loan and firm characteristics. The sample includes all borrowings of a random sample of 750 middle-market firms originated over the 2010-2012 period. Fixed-rate loans are dropped from the performance pricing regressions. All columns include research expenses, PP&E, current ratio, the logarithm of firm age, market-to-book, sales growth, volatility, and past returns as additional controls. To save space, we do not report the coefficients for these variables. Table B1 in the appendix defines the variables. Industry fixed effects are based on Fama-French 12 industries. *t*-statistics adjusted for firm-level clustering are reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

5.7 Table 7: Nonprice terms of bank versus nonbank loans

Read.

- Nonbanks make smaller loans but often with longer maturity in OLS specifications.
- Nonbank loans are less likely to be senior and less likely to be secured.
- Nonbank loans are substantially less likely to contain financial covenants.
- They are less likely to use performance pricing and more likely to include warrants or convertibility.
- They are much more likely to be fixed-rate and more likely to be second lien.

Economic message. Nonbanks are not just banks with a different label. They use a different contracting technology: less covenant-based monitoring and more incentive alignment through equity-like features.

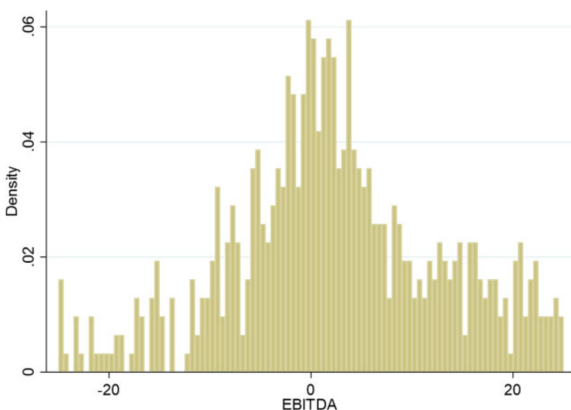


Figure 2
Histogram of trailing 12-month EBITDA

This figure illustrates a histogram of trailing 12-month EBITDA for borrowers with EBITDA in the -\$25 million to \$25 million range. The bin width is \$500,000. The sample includes all borrowings of a random sample of 750 middle-market firms originated and filed with the SEC during the 2010–2015 period.

Table 8
Fuzzy RDD using the coverage error optimal bandwidth selector

	Bandwidth	Left obs.	Right obs.	1st stage	2nd stage
<i>Initial interest rate</i>	25.92	143	177	−0.35*** (−3.69)	589.41*** (3.30)
<i>ln(Amount)</i>	26.25	151	202	−0.34*** (−3.51)	−0.84 (−1.21)
<i>Maturity</i>	31.63	149	224	−0.37*** (−4.19)	−2.91* (−1.79)
<i>Seniority</i>	31.42	154	229	−0.34*** (−3.77)	−0.07 (−0.14)
<i>Security</i>	30.94	153	226	−0.34*** (−3.76)	−0.33 (−1.20)
<i>Second lien</i>	27.91	151	211	−0.34*** (−3.56)	−0.04 (−0.81)
<i>Financial covenants</i>	28.26	151	214	−0.34*** (−3.64)	−0.56* (−1.78)
<i>Performance pricing</i>	36.49	156	252	−0.33*** (−3.82)	−0.05 (−0.10)
<i>Warrants</i>	27.94	150	211	−0.35*** (−3.77)	0.24 (1.08)
<i>Convertible</i>	24.51	147	199	−0.36*** (−3.65)	−0.02 (−0.47)
<i>Fixed-rate loan</i>	30.75	148	218	−0.36*** (−3.97)	0.47* (1.78)
<i>Up-front fee</i>	35.95	142	210	−0.37*** (−4.26)	137.55** (2.50)
<i>Annual fee</i>	29.91	140	190	−0.32*** (−3.38)	26.76 (1.33)
<i>Bankrupt_{t+5}</i>	36.98	156	253	−0.33*** (−3.82)	0.06 (0.32)
<i>Δprofitability_{t+3}</i>	22.75	109	143	−0.30** (−2.33)	0.02 (0.03)
<i>Stock return [1,t+3]</i>	24.96	146	193	−0.36*** (−3.50)	0.12 (1.05)

The table reports the results of fuzzy RDD estimation using local linear polynomials for various outcome variables, as well as cash flow loans only. The treatment is borrowing from a nonbank. The running variable is trailing 12-month EBITDA, with discontinuity at zero. The slope of the effect of the running variable on the probability of treatment is allowed to differ between the left and right of the discontinuity. The estimators are constructed using a triangular kernel. Symmetric bandwidths around zero are determined using the coverage error optimal (CER) bandwidth selector of Calonico, Cattaneo, and Farrell (2018). The CER bandwidth selector depends on the structure of all the data and must be reestimated for each outcome variable. The table reports the bandwidth, the number of observations included to the left and right of the discontinuity, the first-stage effect of an indicator for negative EBITDA on the treatment probability, and the second-stage estimate of the treatment effect on the outcome variables. *z*-statistics using bias-adjusted standard errors from Calonico, Cattaneo, and Farrell (2018) that adjust for clustering at the firm level are reported in parentheses. The following covariates are included, with coefficients omitted for brevity: the logarithm of total assets, leverage, market-to-book, sales growth, R&D, PP&E, cash, receivables, inventory, the logarithm of firm age, volatility, past stock returns, the year of loan origination, and industry effects. The estimation for performance pricing omits fixed-rate loans. **p* < .1; ***p* < .05; ****p* < .01.

5.8 Figure 2 and Table 8: Fuzzy RDD around zero EBITDA

Read.

- The first stage is strong: negative EBITDA lowers the probability of bank borrowing and raises nonbank borrowing.
- The second-stage interest-rate effect is about 589 basis points.
- Nonbank loans near the cutoff are less likely to include financial covenants.

- Nonbank loans are also more likely to be fixed-rate and to charge up-front fees.
- The RDD does not show worse bankruptcy, profitability, or stock-return outcomes for nonbank borrowers.

Economic message. Around a regulatory boundary, the marginal effect of being pushed to a nonbank is expensive debt with different contract terms, but not obviously worse ex post borrower performance.

5.9 Table 9: Probability of bankruptcy for bank versus nonbank loans

Table 9
Probability of bankruptcy for bank versus nonbank loans

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Nonbank</i>	0.077*** (3.33)	0.037 (1.53)	0.011 (0.52)	−0.006 (−0.23)	0.011 (0.28)	0.002 (0.04)
<i>ln(Assets)</i>		0.018** (2.29)	0.025*** (3.07)	0.021** (2.13)	0.019* (1.82)	0.025** (2.02)
<i>EBITDA</i>		−0.000** (−2.54)	−0.000** (−2.51)	−0.000** (−2.14)	−0.000 (−1.09)	−0.000 (−1.11)
<i>EBITDA < 0</i>		0.065* (1.91)	0.049 (1.51)	0.047 (1.32)	0.004 (0.11)	0.001 (0.04)
<i>Leverage</i>		0.105** (2.08)	0.067 (1.47)	0.058 (1.41)	0.198*** (2.92)	0.120** (2.14)
<i>ΔLeverage</i>		0.180** (2.21)	0.120 (1.48)	0.100 (0.93)	0.265** (2.49)	0.225* (1.68)
<i> Asset growth </i>		0.048* (1.83)	0.078*** (2.90)	0.089** (2.47)	0.036 (1.03)	0.048 (1.12)
<i>PP&E</i>		0.056 (0.81)	0.042 (0.61)	0.015 (0.21)	0.003 (0.04)	−0.008 (−0.09)
<i>Current ratio</i>		−0.012** (−2.44)	−0.010** (−2.11)	−0.010* (−1.72)	−0.010* (−1.71)	−0.010 (−1.43)
<i>Research expenses</i>		0.169 (1.35)	0.134 (1.00)	0.143 (0.97)	0.151 (0.99)	0.139 (0.86)
<i>ln(Firm age)</i>		0.010 (0.68)	0.013 (0.94)	0.008 (0.53)	0.015 (0.94)	0.015 (0.89)
<i>Market-to-book</i>			−0.008 (−1.16)	−0.001 (−0.18)	−0.007 (−0.76)	−0.001 (−0.13)
<i>Sales growth</i>			0.018 (0.50)	0.004 (0.09)	−0.022 (−0.50)	−0.034 (−0.74)
<i>Volatility</i>			0.054** (2.39)	0.053** (2.26)	0.028 (1.03)	0.037 (1.28)
<i>Past return</i>			−0.146*** (−5.00)	−0.121*** (−3.67)	−0.146*** (−3.56)	−0.110** (−2.28)
<i>ln(Distance)</i>				−0.001 (−0.25)		−0.005 (−0.86)
<i>Same county</i>				−0.030 (−1.34)		−0.026 (−0.90)
<i>Constant</i>	0.007 (0.25)	−0.171** (−2.40)	−0.220*** (−2.84)	−0.157* (−1.79)	−0.193** (−2.15)	−0.179* (−1.77)
<i>Year effects</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry effects</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,269	1,193	1,121	833	626	478
<i>R²</i>	.08	.14	.19	.17	.21	.16

This table reports the results of linear probability models of borrower's bankruptcy over 5 years after loan origination. There are 101 deals by 52 borrowers that result in bankruptcy within 5 years. In column 5, the sample is limited to nonconvertible first-lien senior secured floating rate loans with financial covenants and without warrants. Table B1 in the appendix defines the variables. Industry fixed effects are based on Fama-French 12 industries. *t*-statistics are adjusted for firm fixed effects. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Read.

- In a raw comparison, nonbank loans are associated with higher bankruptcy probability.
- Once firm characteristics are added, the nonbank coefficient becomes small and statistically insignificant.
- Adding market-based variables, distance controls, and restrictions to comparable loan contracts does not restore a significant nonbank effect.

Economic message. Nonbanks charge more, but the evidence does not imply that they merely select unobservably worse borrowers or that their loans lead to higher default after controlling for borrower risk.

6 Short summary

- The paper studies why publicly traded middle-market firms borrow directly from nonbank lenders.
- The data are hand-collected credit agreements from 2010–2015, covering 1,269 loans by 579 borrowers.
- About one-third of loans are directly originated by nonbanks, and many are missing from DealScan.
- The core empirical result is that negative EBITDA and debt/EBITDA greater than six strongly predict nonbank borrowing.
- The mechanism is bank regulation: banks face supervisory costs when lending to unprofitable or highly levered borrowers.
- The regulatory interpretation is supported by evidence from OCC-supervised banks, local OCC market share, cash-flow versus asset-based loans, and a fuzzy RDD around zero EBITDA.
- Nonbank loans carry much higher interest rates, especially for negative EBITDA borrowers and after regulatory tightening.
- Nonbank contracts differ from bank contracts: fewer financial covenants, fewer performance-pricing provisions, more warrants, more convertibility, more fixed-rate loans, and more second-lien positions.
- Ex post performance tests do not show that nonbank borrowers are worse after conditioning on observables.
- The economic message is that bank regulation can shift credit provision outside the banking sector and generate market segmentation with higher borrowing costs for constrained firms.

7 Conclusion

7.1 Core conclusion

Chernenko, Erel, and Prilmeier show that direct nonbank lending to middle-market firms is not a marginal phenomenon. It accounts for about one-third of observed loans in their hand-collected sample. The central driver is bank regulation, especially rules and supervisory practices that make banks reluctant to lend to borrowers with negative EBITDA or very high debt/EBITDA.

7.2 Economic intuition

The economic logic is search-based market segmentation. A borrower with negative EBITDA knows that many banks may decline the loan or demand restrictive terms because of regulatory scrutiny. Once the borrower turns to nonbanks, the lender knows the borrower has fewer outside options. This creates bargaining power for the nonbank and helps explain high interest rates. The mechanism is strongest for cash-flow loans and small firms because these borrowers are most dependent on local bank credit.

7.3 Takeaway

The paper's main lesson is that regulation can change not only how much credit banks supply but also who supplies credit, which firms are pushed into alternative credit markets, and what loan contracts those firms receive. Nonbanks fill a credit gap, but they do so with higher prices and different monitoring and incentive tools.